

Spin-Charge Conversion

SCC enables:

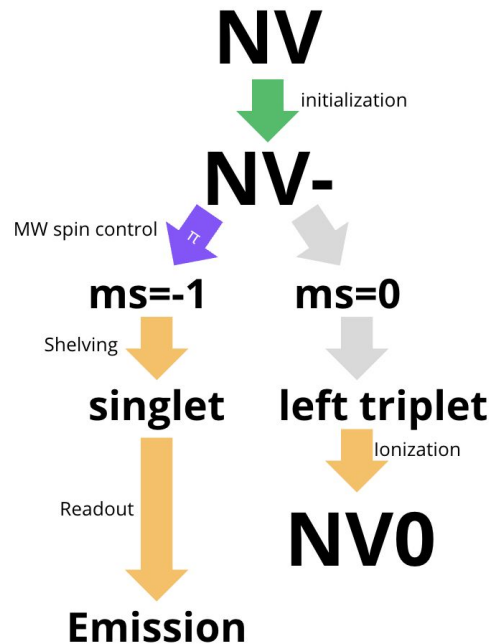
- extended measurement times

- increasing precision

- minimizing noise in the readout compared to the commonly used fluorescence detection

SCC utilizes the intersystem crossing dynamics of the NV-excited state to protect one spin state from an intense ionization pulse, leading to a spin-dependent charge distribution.

A subsequent charge-selective optical pulse detects the resulting distribution



Intersystem crossing

When an electron in a molecule with a singlet ground state is excited (*via* absorption of radiation) to a higher energy level, either an excited singlet state or an excited triplet state will form.

Singlet state is a molecular **electronic state** such that all **electron spins** are paired. That is, the spin of the excited electron is still paired with the **ground state** electron (a pair of electrons in the same energy level must have opposite spins, per the **Pauli exclusion principle**).

In a **triplet state** the excited electron is no longer paired with the ground state electron; that is, they are parallel (same spin). Since excitation to a triplet state involves an additional "forbidden" spin transition, it is less probable that a triplet state will form when the molecule absorbs radiation.

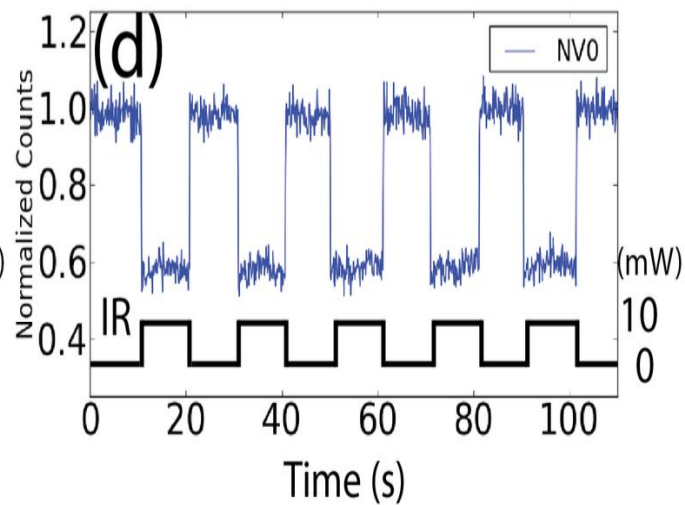
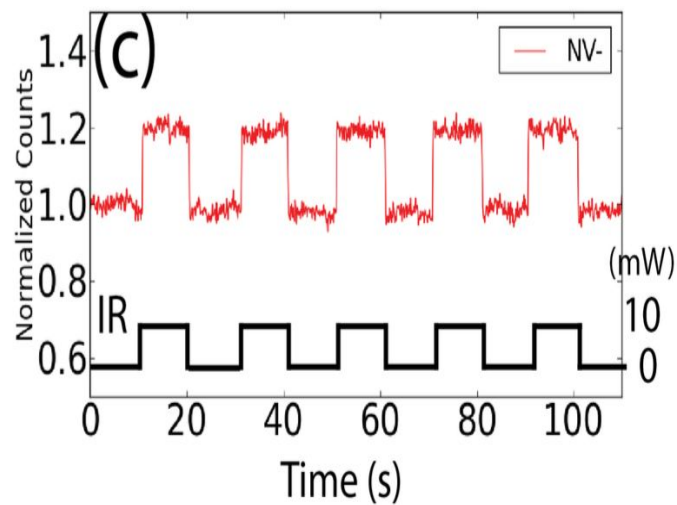
When a singlet state nonradiatively passes to a triplet state, or conversely a triplet transitions to a singlet, that process is known as intersystem crossing. In essence, **the spin of the excited electron is reversed**. The probability of this process occurring is more favorable when the vibrational levels of the two excited states overlap, since little or no energy must be gained or lost in the transition.

The time scale of intersystem crossing is on the order of 10^{-8} to 10^{-3} s, one of the slowest forms of relaxation.

(wikipedia)

Peng's work on SCC

- NV0 converts to NV- under IR(1064 nm) laser excitation
- Slow charge transfer between NV- and NV0 charge state
- Charge transfer mediated by 1064 nm laser is spin-dependent

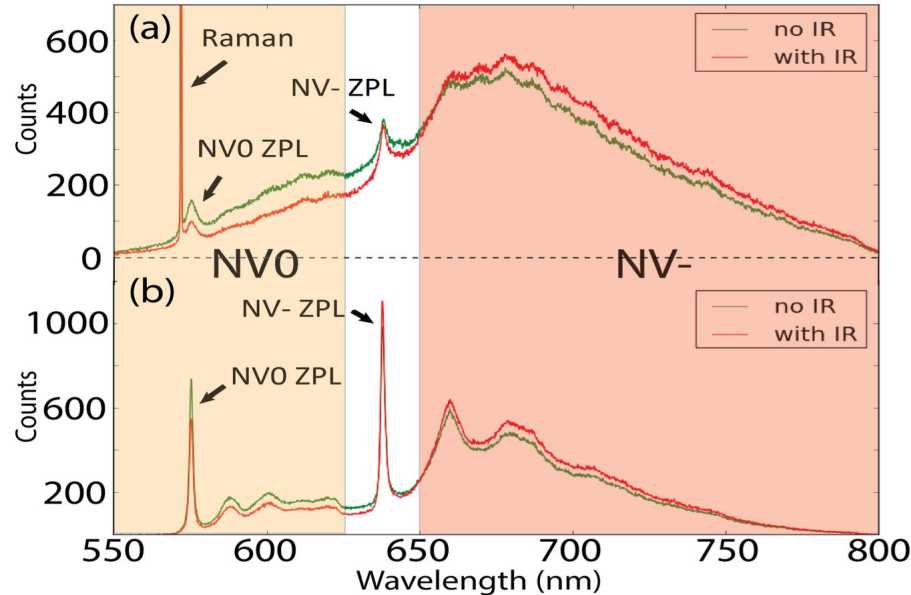


(c) NV⁻ PL increases with IR

(d) NV⁰ PL decreases with IR

Evidence for NV0 converting to NV-

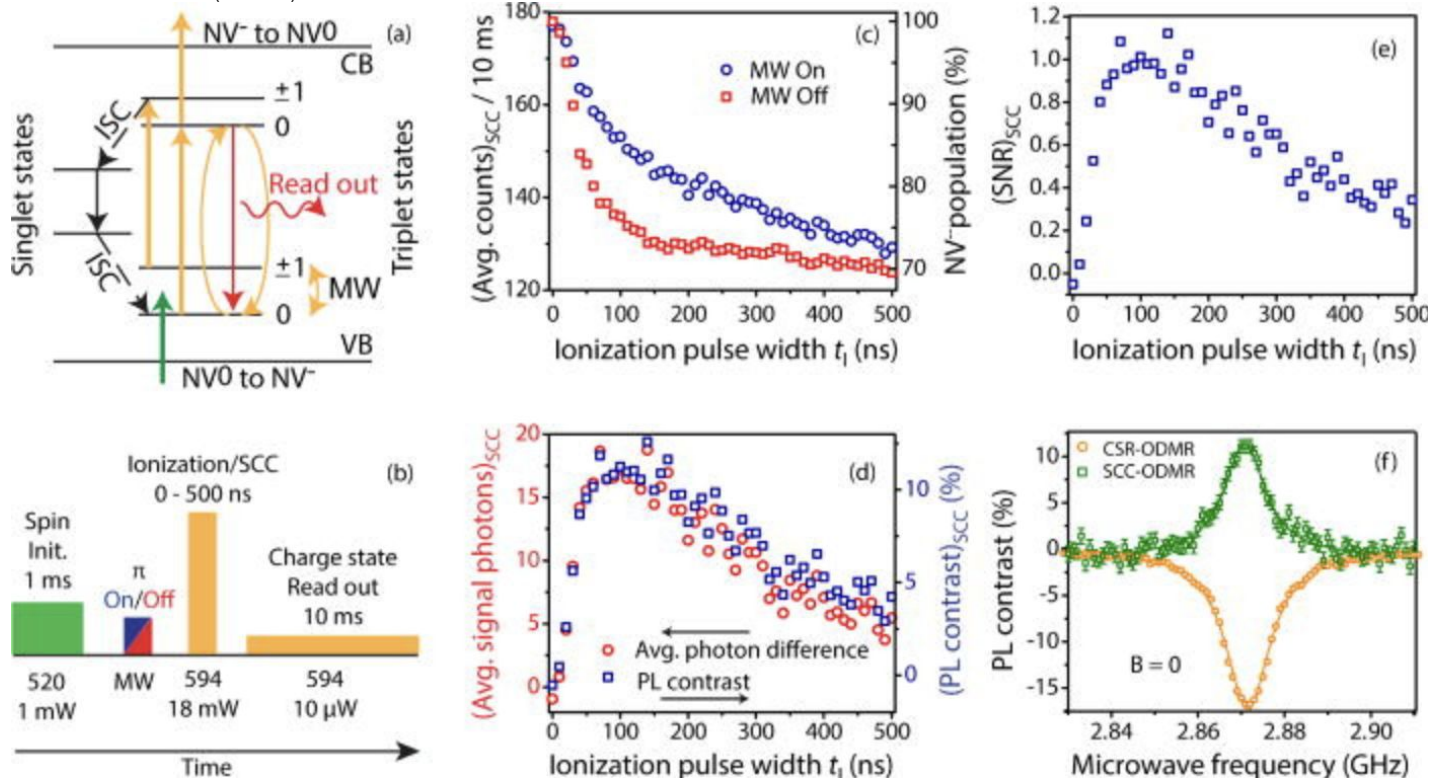
Enhanced Photoluminescence(PL) from NV-



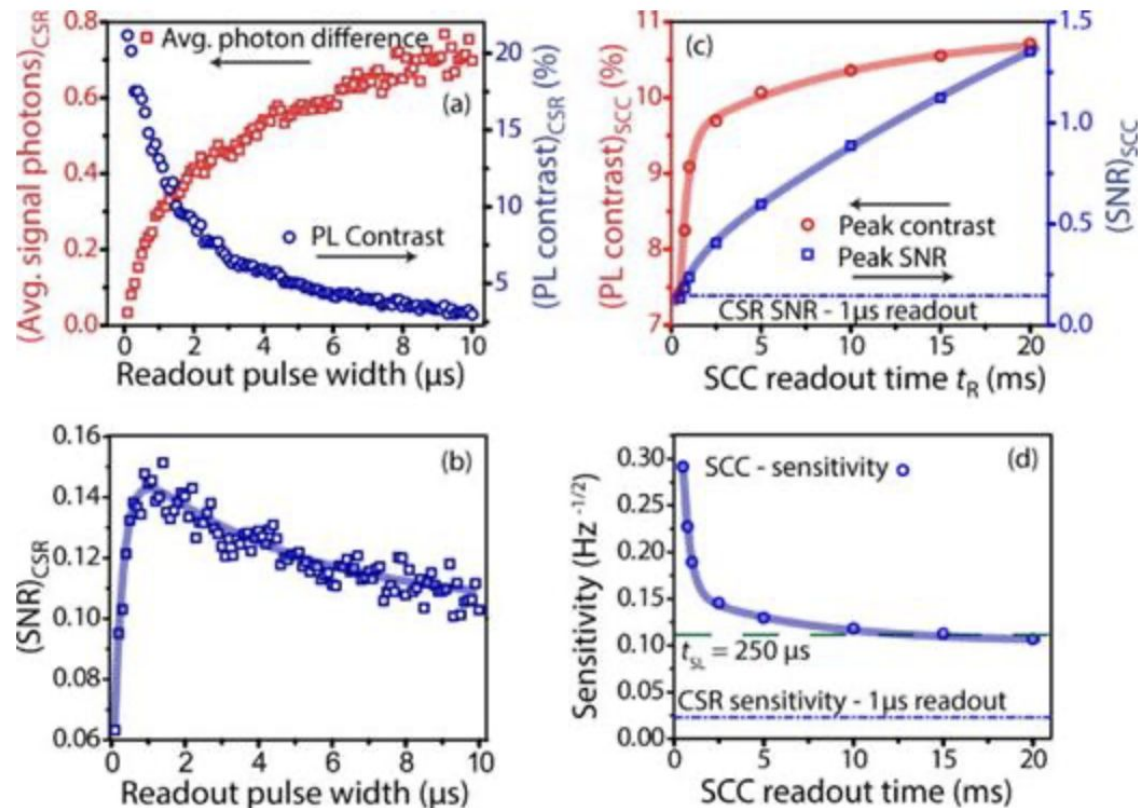
- (a) Bulk Diamond (NV ensemble)
- (b) Nanodiamond (NV ensemble)

Bulk diamond: Ensemble:

<https://pubs.aip.org/aip/apl/article/113/12/122404/986214/Spin-readout-via-spin-to-charge-conversion-in-bulk>
a tenfold reduction (or better) in measurement time in SCC than CSR



Fundamentals of SCC. (a) NV⁻ energy level diagram and schematics of spin-to-charge conversion. Charge and spin initialization is carried out via 520 nm laser excitation (green arrow). We ionize and readout the NV charge state via strong and weak 594 nm laser light (thick and thin orange arrows, respectively). The wavy, red arrow denotes emitted photons during detection. MW: microwave, ISC: inter-system crossing, CB: conduction band, and VB: valence band. (b) Pulse sequence for the SCC-based spin-state measurement. (c) Average number of photons acquired by the 10 ms readout pulse after the spin-to-charge conversion, with and without the MW π-pulse. (d) Average number of signal photons and contrast for a 10 ms readout pulse for varying ionization pulse widths. (e) Signal-to-noise ratio of the SCC measurement for varying ionization pulse widths. (f) Comparison of the pulsed ODMR signals acquired with spin-to-charge conversion (SCC) and conventional spin readouts (CSR) at zero external magnetic field ($B = 0$ G). Solid lines are Lorentzian peak function fits to the data, and the error bars are for the photon shot noise.

FIG. 2.

Comparison between SCC and the standard spin readout. (a) Average number of signal photons and contrast and (b) SNR for the conventional 520 nm spin state readout without spin-to-charge conversion, as a function of the readout pulse width. (c) PL contrast (SNR) for the SCC readout of variable duration (red and blue traces, left and right vertical scales, respectively); for comparison, the plot includes the SNR from the 520 nm, 1 μs readout (dash-dotted blue line). (d) A similar comparison for the readout sensitivity.

noise ratio (SNR). A lower photon count (and hence an increased ionization rate) is observed for $m_S = 0$ (when the MW is off) due to spin-to-charge conversion [Fig. 1(c)]. About 25% (14%) of NVs are converted to NV^0 within the first 100 ns of the ionization pulse for $m_S = 0$ ($m_S = \pm 1$). From here, we define the average signal photon (SP) count as

$$SP = C_{MW\ On} - C_{MW\ Off},$$

where $C_{MW\ On, Off}$ are the average counts shown in Fig. 1(c). The SCC photo-luminescence (PL) contrast is given by

$$PL\ Contrast = 1 - \frac{C_{MW\ Off}}{C_{MW\ On}}.$$

The SNR is calculated with the photon shot noise as

$$SNR = \frac{SP}{\sqrt{(C_{MW\ On} + C_{MW\ Off})}}.$$

0.2 mm thick, type 1b diamond with a nominal nitrogen concentration of 1 ppm purchased from Delaware Diamond Knives. NV depth of about 20 μm from the sample surface via a home-built laser-scanning microscope with a 1.3 NA oil immersion objective; the beam diameter is ~ 300 nm corresponding to a detection volume V_d of approximately $0.07 \mu m^3$. From the observed fluorescence, we conclude that the number of NVs within V_d amounts to ~ 30 , hence yielding an NV concentration of ~ 3 ppb. Ionization 18 mW 594 nm.

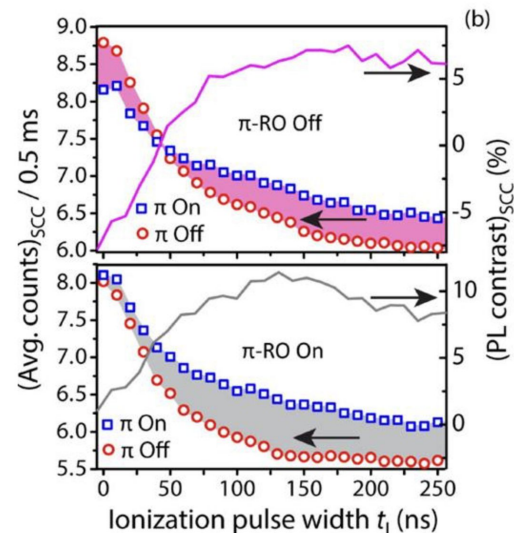
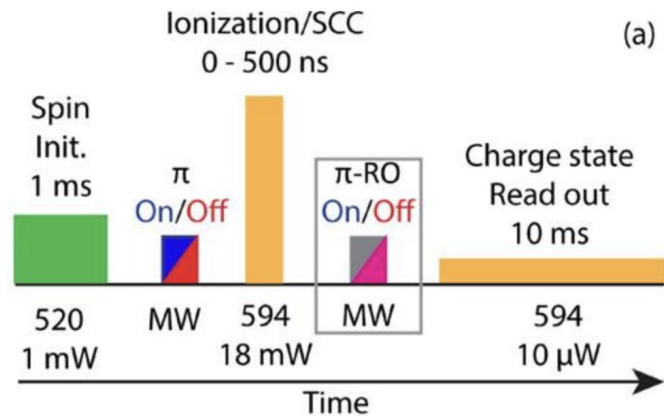
The pulse sequence used for the experimental realization of SCC [Fig. 1(b)] consists of a 520 nm initialization laser pulse (1 mW, 1 ms), a MW π -pulse for spin inversion, an 18 mW, 594 nm, laser ionization pulse of varying durations, and a weaker 594 nm readout pulse (10 μW , 10 ms).

the first step of the ionization process, the electron is taken to the excited state where it has a comparatively longer radiative lifetime of 13 ns and thus a greater, laser-power-dependent probability of being ionized by a second photon. On the other hand, if the initial spin projection is $m_S = \pm 1$, the excited state rapidly undergoes intersystem crossing and is shelved for 300 ns in the ground singlet, a state with a low two-photon ionization cross section. Thus, the efficiency of the spin-to-charge conversion depends on (i) the spin selectivity of the intersystem crossing, (ii) the ionization probabilities of the triplet and singlet states, (iii) the ionization laser power and pulse width, and (iv) the recombination probability.

the SCC protocol preferentially ionizes NVs whose spin state immediately prior to the ionization pulse is $m_s=0$. At short readout times, less than optimum contrast because NVs initially in the $m_s=+1$ states must be spin-polarized before the fluorescence reaches a maximum. This problem can be prevented through the addition of a second MW pulse transferring the populations into the $m_s=0$ state immediately before the readout [Fig. 3(a)].

This crossing between PL traces with different initial spin states effectively reduces the contrast and hence negatively impacts the resulting SNR. As shown in the lower half of Fig. 3(b), the addition of a π -RO pulse immediately prior to the readout pulse removes this problem, allowing us to reach a contrast of 12%, on par with the contrast achieved with longer readout times.

FIG. 3.



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Improving SCC contrast with a readout π -pulse (π -RO). (a) Pulse sequence for comparing readouts with and without a π -RO (highlighted with a dotted square). (b) Gray (pink) shaded regions bound by the average counts of 0.5 ms SCC readout with (without) MW π -RO.

sensitivity

η of the spin state measurement defined as

$$\eta = \frac{\sqrt{t_T}}{\text{SNR}},$$

where the total protocol time t_T is here given by

$$t_T = t_I + t_{SL} + t_R,$$

$$\text{Speedup Factor} = \left(\frac{\eta_{\text{CSR}}}{\eta_{\text{SCC}}} \right)^2,$$

$$\text{Readout Time} = T_0 \left(\frac{P_0}{P} \right)^2,$$

where $T_0 = 20$ ms and $P_0 = 10$ μW denote the readout time and laser power used herein,

Nanodiamond: Ensembles

<https://pubs.acs.org/doi/10.1021/acsnano.8b01265>
(20.21)(22)

Sizes ranging between 20 and 70 nm

nanodiamonds milled from type-Ib high-pressure, high-temperature (HPHT) diamond

Figure 1. Charge and spin control of individual nanodiamonds. (a) Experimental setup consisting of a two-color home-built confocal microscope imaging nanodiamonds dispersed on a microwave antenna device fabricated on silicon (SPAD, single-photon avalanche diode; AOM, acousto-optic modulator; NDs, nanodiamonds; MWs, microwaves). Inset: Experimental demonstration of a T_1 spin relaxometry measurement using PL (blue) and SCC (red) depicting the SNR amplification for a nanodiamond agglomerate. Atomic force microscope measurements (b) and PL images (c) of the region boxed in (d) indicate the presence of individual nanodiamonds.

Figure 1

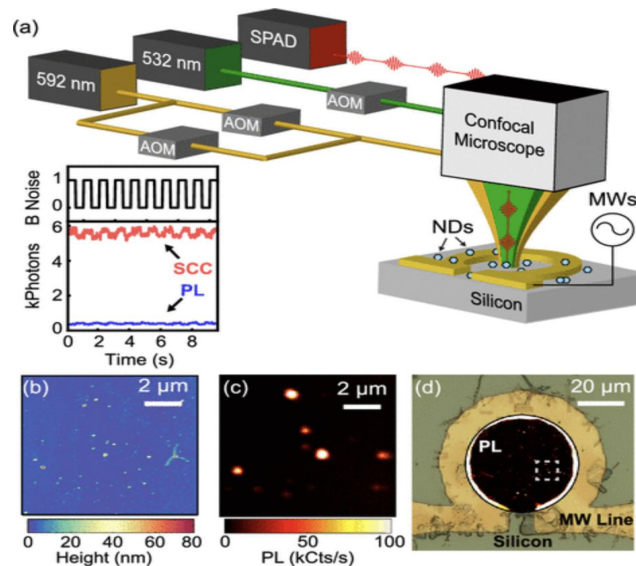
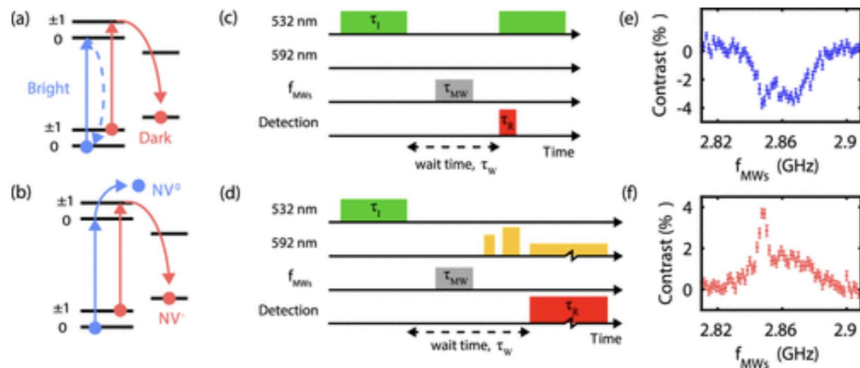


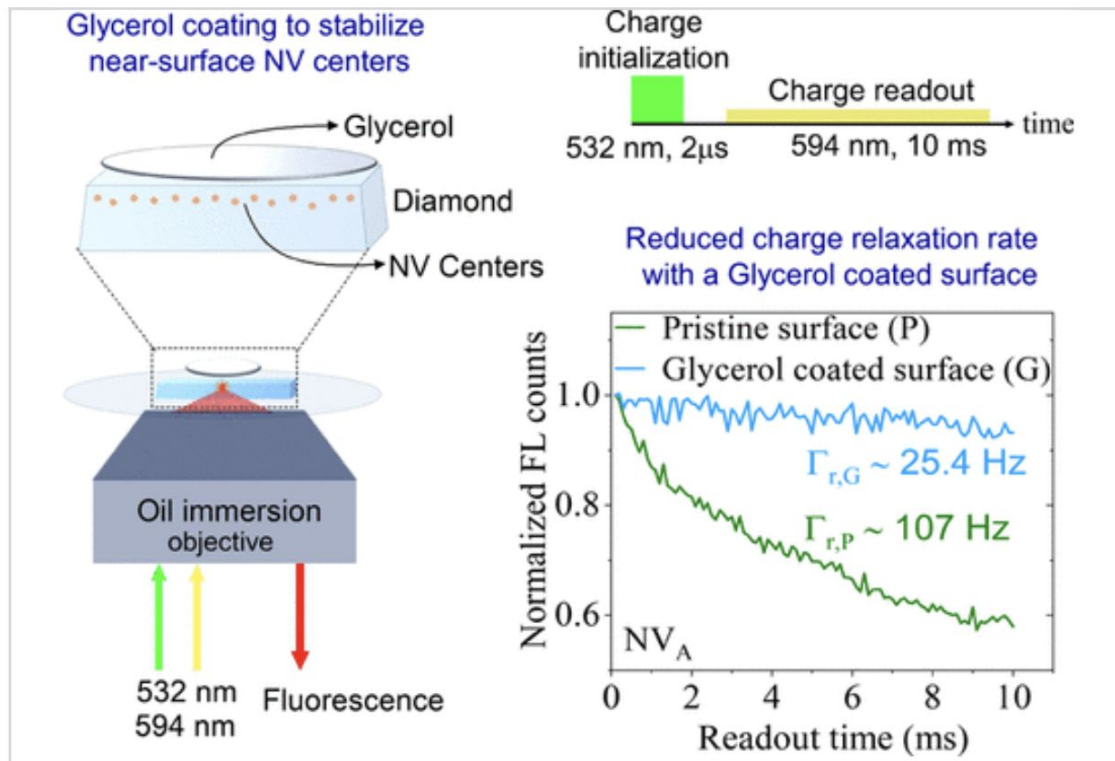
Figure 4

Figure 4. Spin-to-charge conversion in nanodiamonds. (a,b) Diagrams of PL spin readout and the SCC process, respectively, for NV⁻. Solid lines represent pumped transitions, and dotted lines represent radiative transitions. (c,d) Measurement timing diagrams for performing pulsed electron spin resonance (ESR) using PL or SCC readout, respectively. f_{MW} corresponds to the variable frequency of the microwave pulse, and the break in the readout pulse of (d) indicates a large span of time. τ_i , initialization time; τ_{MW} , microwave pulse time; τ_r , readout duration. (e,f) Measured ESR spectra for a single nanodiamond using PL and SCC protocols, respectively.

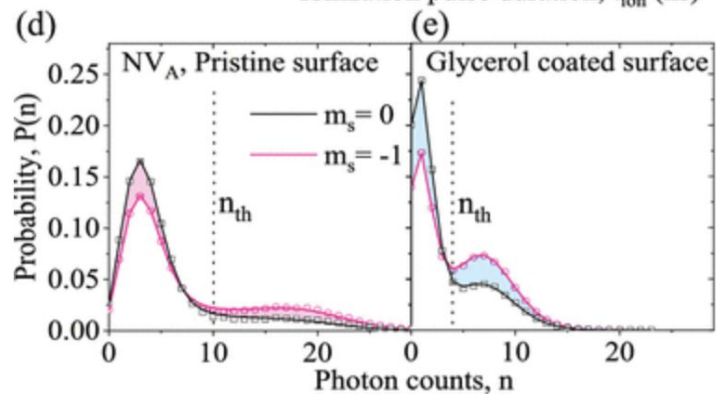
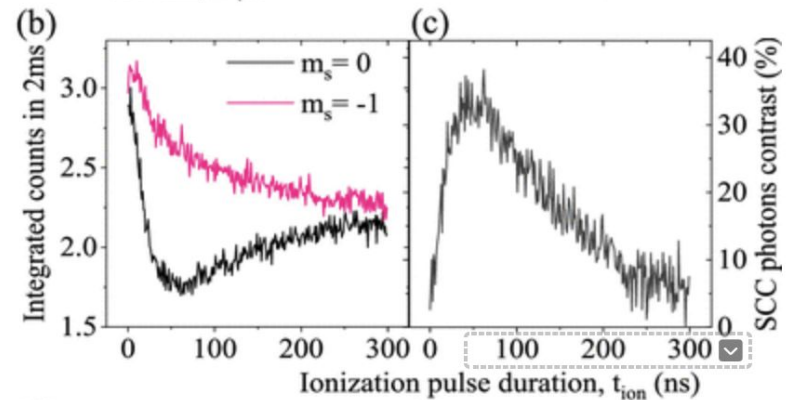
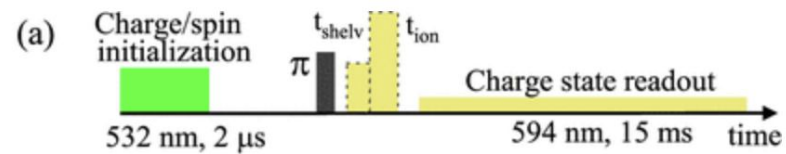


Single NV (532 nm and 594 nm):

<https://pubs.acs.org/doi/10.1021/acsaelm.3c01141#:~:text=Spin%2Dbased%20applications%20of%20the,by%20a%20subsequent%20charge%20readout>



Fluorescence evolution of NV_A during the readout pulse, showing a reduced relaxation rate after adding glycerol ($\Gamma_{r,G}$) as compared to the relaxation with the pristine surface ($\Gamma_{r,P}$). (e, f) Photon-number distribution $P(n)$ of NV_A with the pristine (e) and the glycerol-coated (f) surface. The gray and red-shaded areas illustrate the respective contributions of NV_0 and NV^- .



Single NV (532 nm, 594 nm, and 638 nm lasers):

<https://journals.aps.org/prl/abstract/10.1103/PhysRevLett.114.136402>

Images

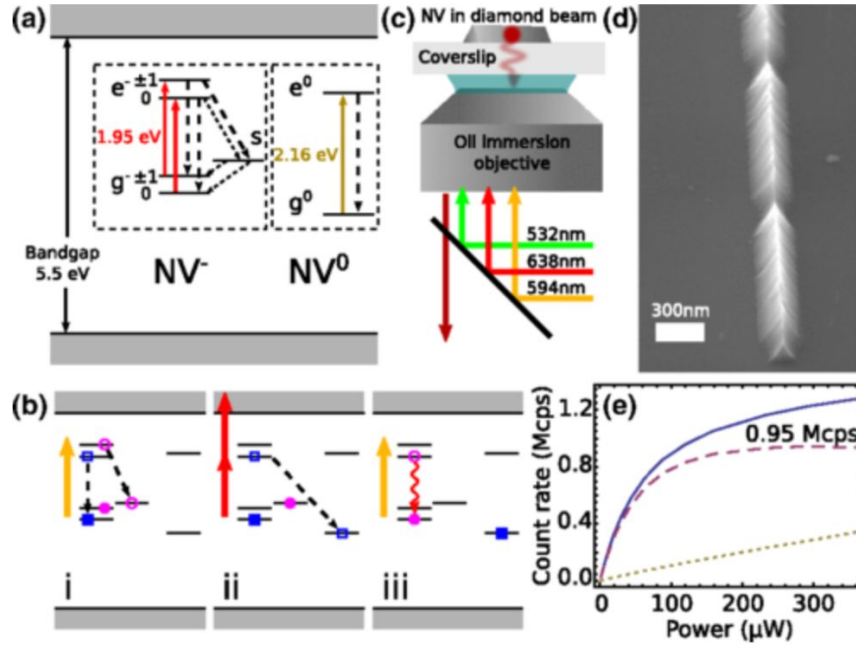


Figure 1

SCC measurement idea. (a) NV center level diagram, indicating triplet ground (g^-), excited (e^-), and metastable singlet (s) states of NV^- and ground (g^0) and excited (e^0) states of NV^0 . (b) SCC measurement process for an initial state of $m_s = 0$ (blue squares) or $m_s = 1$ (pink circles). (i) A 594-nm pulse either shelves into the singlet state ($m_s = 1$) or cycles ($m_s = 0$). (ii) A 638-nm pulse rapidly ionizes the NV^- triplet states to NV^0 . (iii) Single-shot charge-state measurement with 594-nm light. (c) Setup for high collection efficiency from nanobeams. (d) SEM micrograph of a nanobeam transferred to silicon (60° tilt). (e) Saturation fluorescence measurement. Total fluorescence (blue line), background (gold dotted line), and NV signal (red dashed line). The maximum count rate is 0.945 Mcps after background subtraction.

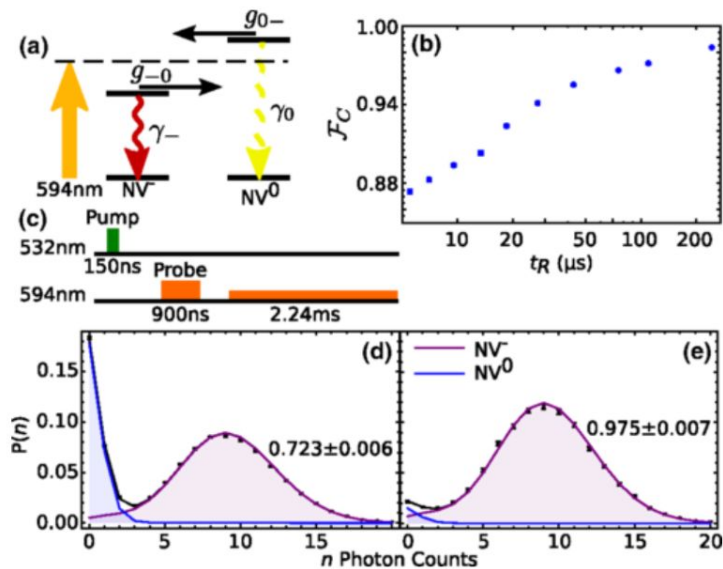


Fig. 2.

Charge-state initialization and readout. (a) Level diagram for charge readout. 594-nm light efficiently excites NV⁻, while weakly exciting NV⁰, yielding photon count rates γ_- and γ_0 , respectively. Ionization occurs from the excited states at rates g_{-0} and g_{0-} . (b) Optimized charge-state readout fidelity $\mathcal{F}_C(t_R)$, determined by measuring γ_- , γ_0 , g_{-0} , and g_{0-} at various illumination intensities. (c) Pulse sequence for charge initialization. A pump-probe sequence initializes into NV⁻ (see the text), and a $t_R = 2.24\text{ms}$ readout (594 nm, 820 nW) verifies the charge state. (d) The readout photon number distribution for 100 000 iterations indicates an initialization fidelity of 0.723 ± 0.006 . The longer-than-optimal t_R ensures an accurate fit of the populations, but here we plot the distribution for the first 240 μs for clarity. (e) Conditioning on one or more probe photons, the fidelity increases to 0.975 ± 0.007 . The black line is a fit to the full 2.24-ms readout. The blue and purple lines indicate NV⁰ and NV⁻ contributions, respectively.

Images

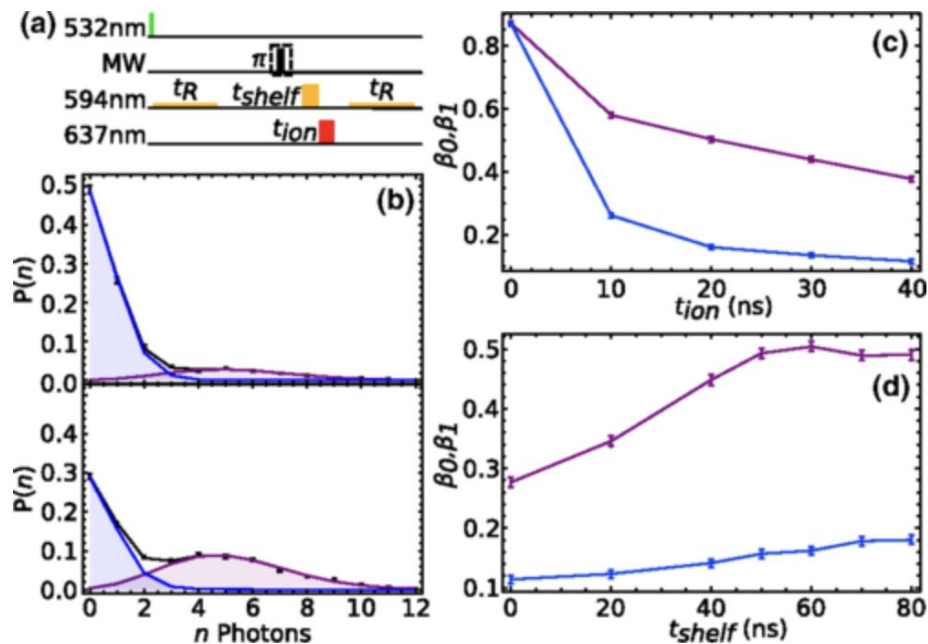


Figure 3

Spin-to-charge conversion. (a) Pulse sequence consisting of initialization (40-ns, 300- μ W pump; 500- μ s, 500-nW probe), followed by microwave (MW) manipulation to prepare in $m_s = 0$ or $m_s = 1$, two-pulse SCC for duration t_{shelf} and t_{ion} , and finally charge readout (500 μ s, 500 nW). (b) Photon number distributions for $t_{shelf} = 60$ ns and $t_{ion} = 20$ ns. An initial $m_s = 0$ state ionizes to NV^0 (top, $\beta_0 = 0.162 \pm 0.007$) while $m_s = 1$ is shelved into the singlet state and protected from ionization (bottom, $\beta_1 = 0.504 \pm 0.009$). (c) Final NV^- population for t_{ion} between 0 and 40 ns ($t_{shelf} = 60$ ns). (d) Final NV^- population for t_{shelf} between 0 and 80 ns ($t_{ion} = 20$ ns), showing dynamics of the shelving process.

Images

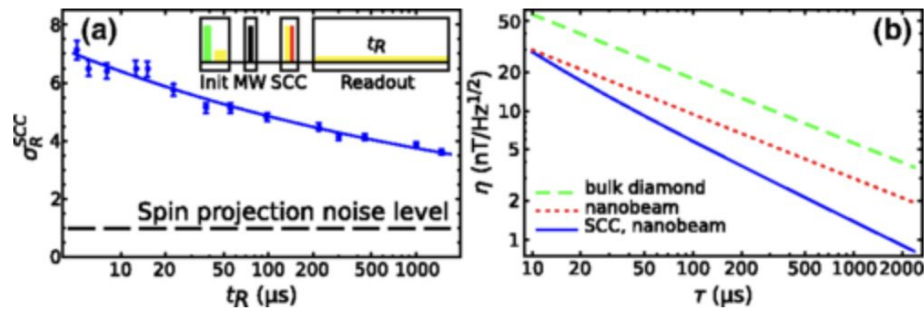


Figure 4

Time dependence and magnetometer sensitivity. (a) Measurement of a $\sigma_R^{\text{SCC}}(t_R)$ (blue points) power law fit (solid line). The inset shows the pulse sequence, consisting of initialization (150-ns, 300- μW pump; 900-ns, 11- μW probe), microwave pulse to prepare $m_s = 0$ or $m_s = 1$, SCC sequence (50-ns, 135- μW shelving pulse; 30-ns, 7.1- mW ionization pulse), and readout. (b) Based on the fit in (a), we directly calculate the magnetometer sensitivity [Eq. (2)] for SCC readout (blue solid line). The sensitivity for conventional readout in bulk diamond ($\sigma_R = 20$, green dashed curve) and diamond nanobeams ($\sigma_R = 10.6$, red dotted curve) is shown for comparison.

Single NV using NIR LASER:

<https://www.nature.com/articles/s41467-021-21781-5>

Fig. 1: Single-shot readout scheme based on SCC.

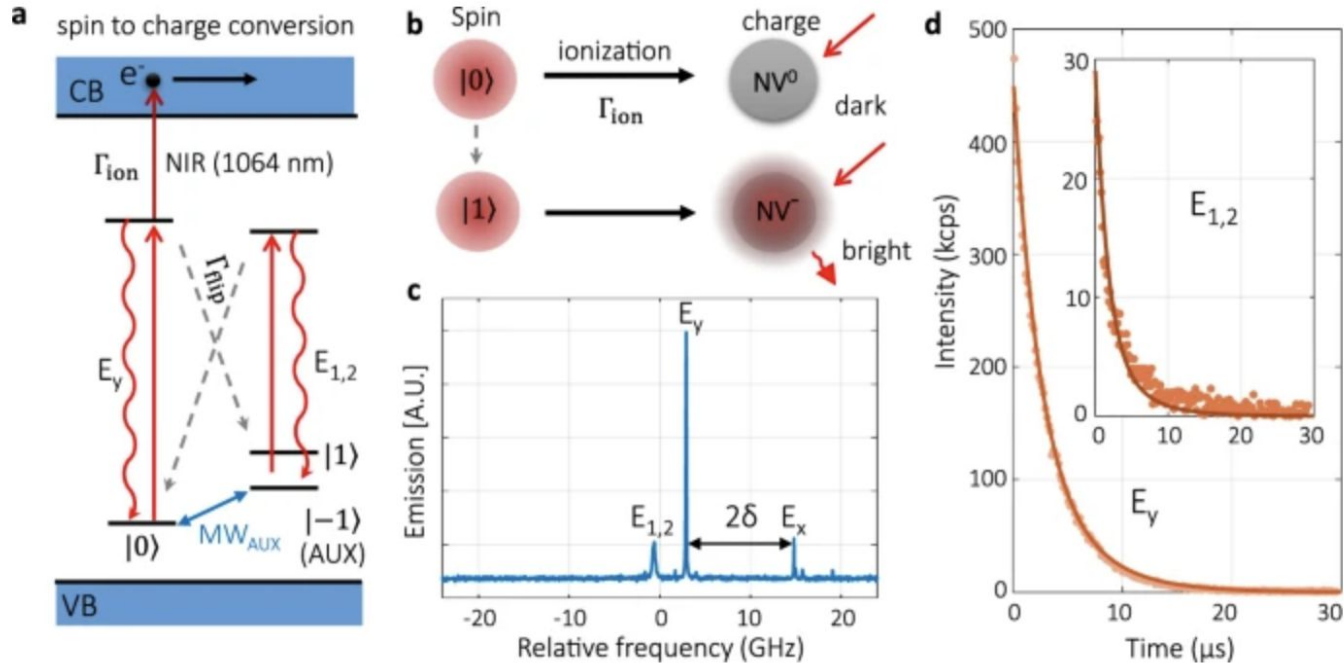


Fig. 2: Non-demolition readout of charge state and ionization rate of NIR light.

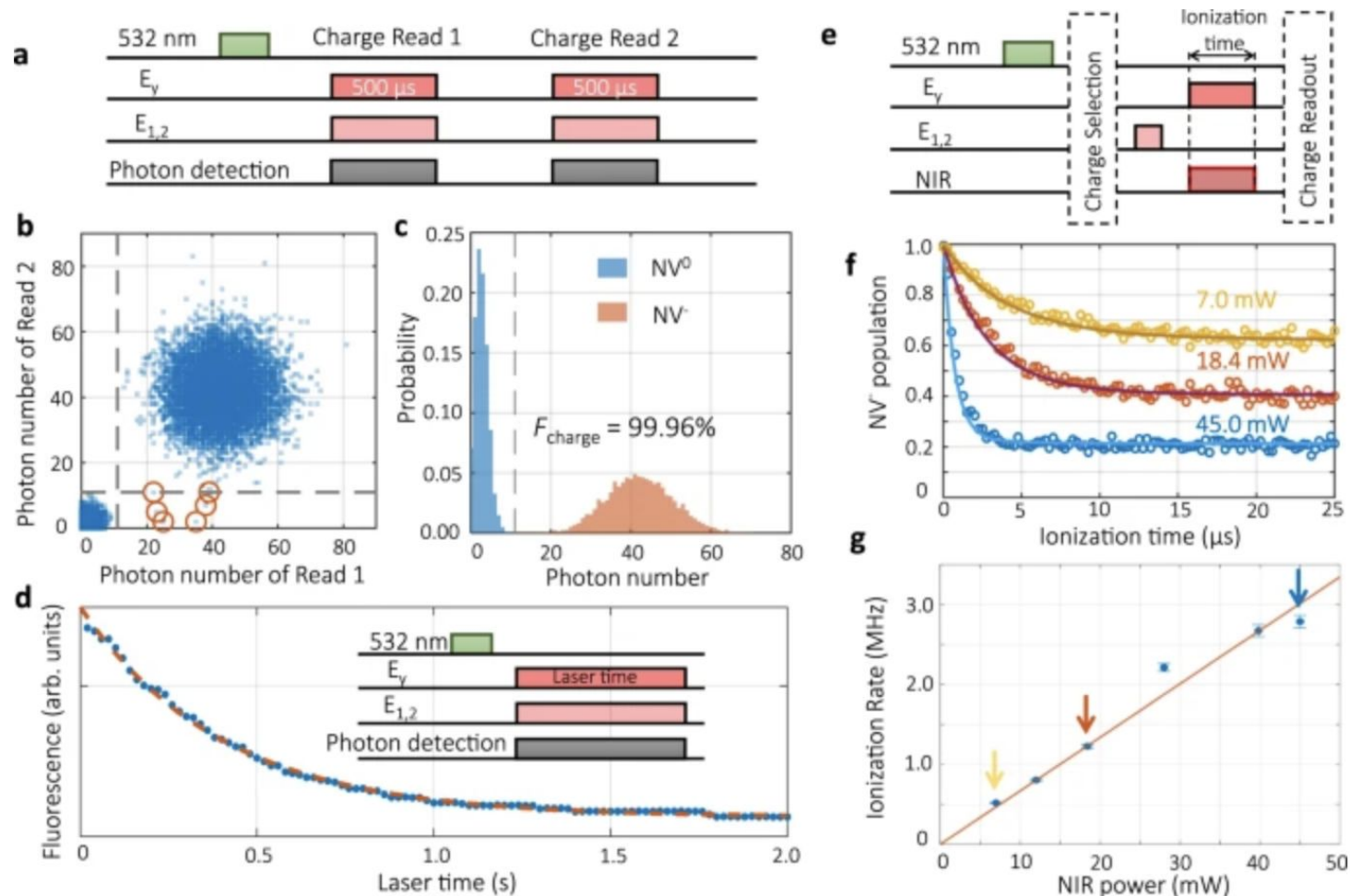


Fig. 3: Single-shot readout of NV electron spin state via SCC.

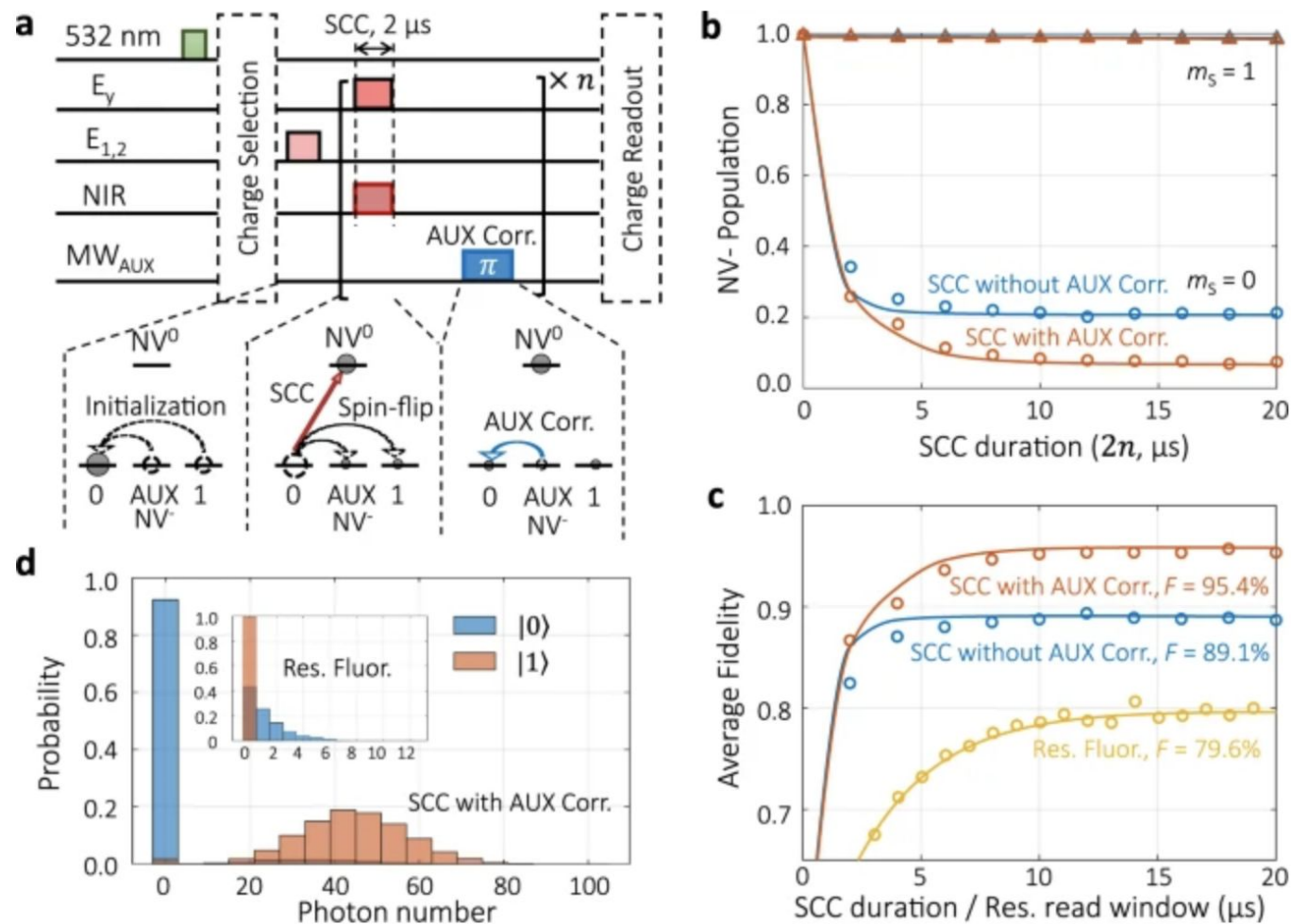
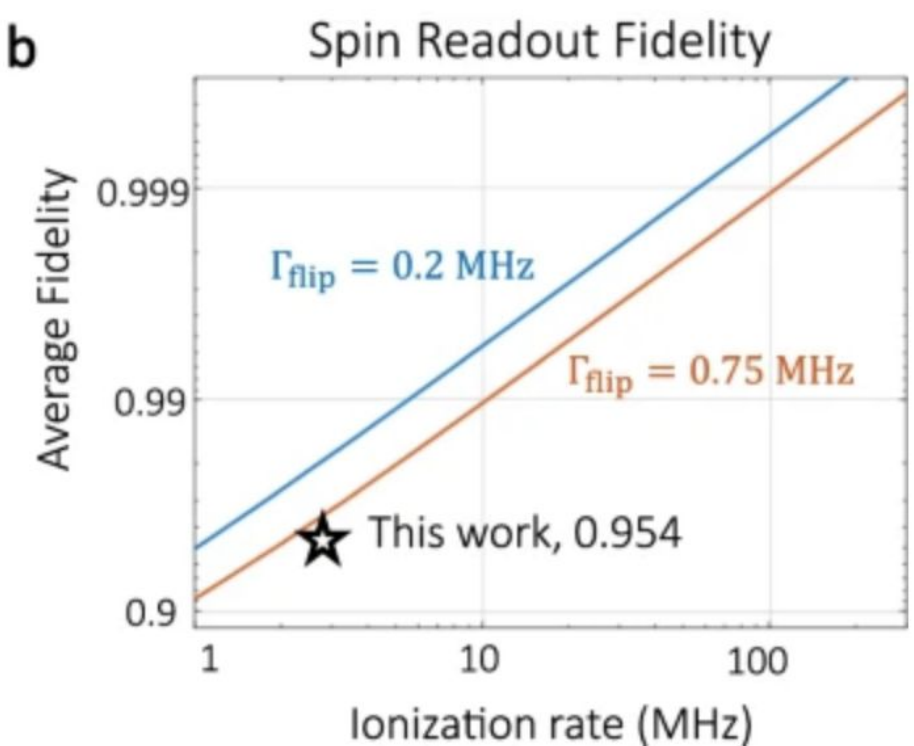
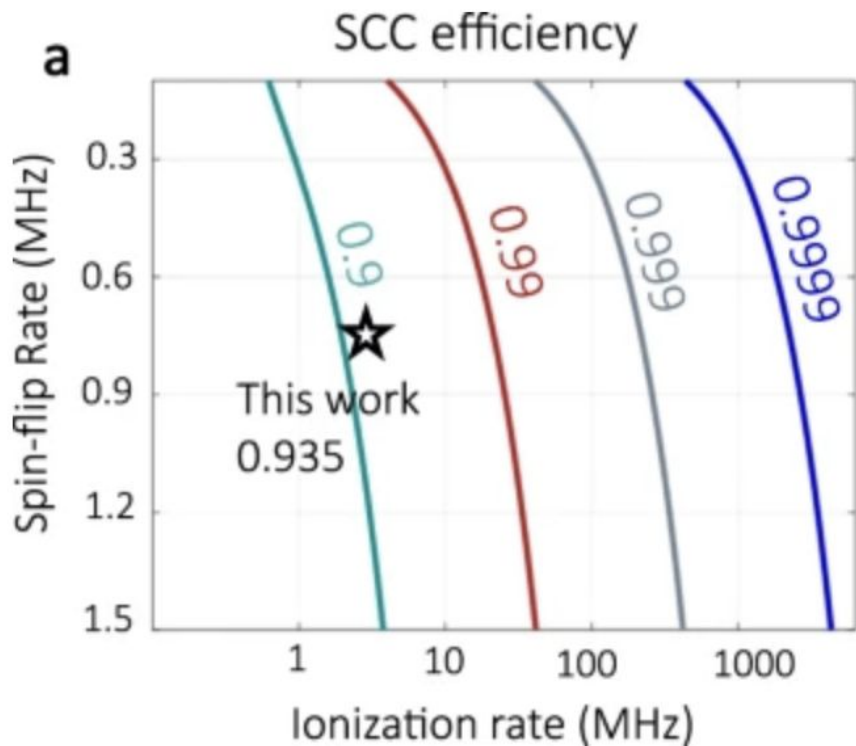


Fig. 4: Exceeding the 99.9% fault-tolerant threshold.



Ensemble (520 nm)

<https://journals.aps.org/prb/abstract/10.1103/PhysRevB.108.075411>

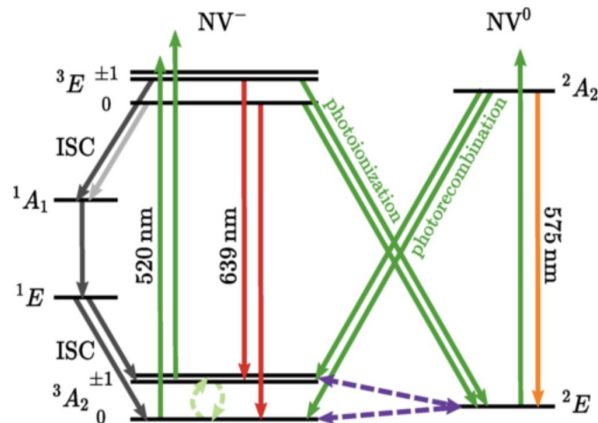
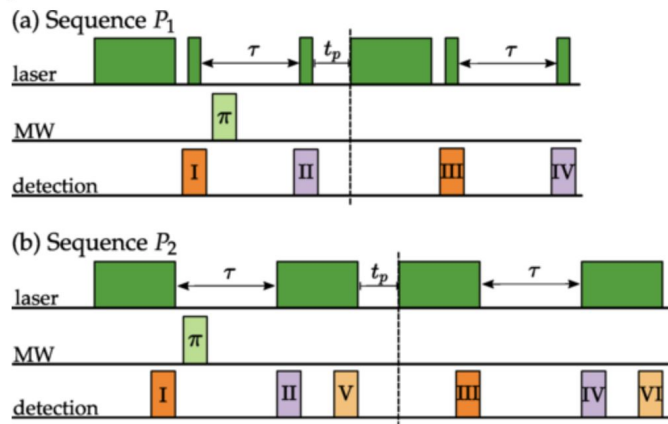


Figure 5

Longitudinal spin relaxation time (T_1) measurement schemes applied in this paper.

The beginning of the second half of each sequence is indicated by a dashed line.

(a) Sequence P_1 . The NV ensemble is spin polarized by a laser pulse. Next, the fluorescence is detected by a control pulse (orange). Within the variable relaxation time τ , a resonant π pulse is applied (light green). The spin state is read out by a third laser pulse (purple). The sequence is repeated with the π pulse omitted after a pause time t_p .

(b) Sequence P_2 . As opposed to P_1 , the readout pulse has the same length as the spin-polarization pulse. The control collection windows III within the initialization pulse and VI within the readout pulse will be compared in this paper.

Figure 1

Level scheme of the NV center in diamond. Depicted are levels of the negatively charged NV⁻ and the neutrally charged NV⁰ and transitions between the two charge states. Gray arrows show transitions between NV⁻'s triplet and singlet states, mediated via intersystem crossing (ISC). Green arrows denote transitions driven by a green laser, red and orange arrows mark the fluorescence of the NV charge states. Light-green dashed arrows denote transitions driven by microwave radiation at 2.87 GHz at zero magnetic field. Additionally, the light-green dashed arrows represent the relaxation of the spin-polarized state to a thermally mixed state without illumination (T_1). The purple dashed arrows denote charge transfer processes in the dark.

<https://pubs.acs.org/doi/10.1021/acs.nanolett.8b01739>

More research

Incorporating **electrodes or surface engineering** to stabilize $NV^- \rightarrow NV^0$ transitions is still an open area.

Research could explore:

- Patterned electrodes to control local electric fields
- Surface treatments that improve SCC contrast without degrading coherence
- Integration with microstructures (waveguides, cavities) to improve photon collection during SCC
- Hybrid readout methods combining **SCC with resonant optical readout** or **microwave-assisted ionization** are not fully explored.
- Could potentially push **faster, near-single-shot readout** in ensembles, even in bulk diamond.